HTD-Mamba: Efficient Hyperspectral Target Detection With Pyramid State Space Model

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***Abstract*— Hyperspectral target detection (HTD) identifies objects of interest from complex backgrounds at the pixel level, playing a vital role in Earth observation. However, the limited target priors constrain the ability to obtain sufficient features or patterns for background-target discrimination, and spectral variation further exacerbates the difficulty of achieving reliable and robust performance. To address these challenges, this article proposes an efficient self-supervised HTD method with a pyramid state space model (SSM), named HTD-Mamba, which employs spectrally contrastive learning to distinguish between target and background based on the similarity measurement of intrinsic features. Specifically, to obtain sufficient training samples and leverage spatial contextual information, we propose a spatial-encoded spectral augmentation (SESA) technique that encodes all surrounding pixels within a patch into a transformed view of the center pixel. In addition, to explore global band correlations, we divide pixels into continuous group-wise spectral embeddings and introduce Mamba to HTD for the first time to model long-range dependencies of the spectral sequence with linear complexity. Furthermore, to alleviate spectral variation and enhance robust representation, we propose a pyramid SSM as a backbone to capture and fuse multiresolution spectral-wise intrinsic features. Extensive experiments conducted on four public datasets demonstrate that the proposed method outperforms state-of-the-art methods in both quantitative and qualitative evaluations. The code is available at https://github.com/shendb2022/HTD-Mamba.**

***Index Terms*— Contrastive learning, hyperspectral target detection (HTD), multiresolution features, spectral variation, state space model (SSM).**

I. Introduction

HYPERSPECTRAL imaging sensors capture contiguous spectral profiles of surface materials, spanning from the ultraviolet to the shortwave infrared bands. By scanning across a scene, each spectral narrowband yields a 2-D geometric image. The composite of these images forms a 3-D data cube, known as the hyperspectral image (HSI). In this cube, the two spatial dimensions depict surface structure, while the third dimension characterizes spectral information. The distinct spectral signatures, arising from variations in absorption, reflection, and scattering across different materials, serve as unique identifiers akin to fingerprints for precise recognition of ground objects [1], [2]. As a result, HSIs are extensively used in applications such as military reconnaissance [3], medical diagnostics [4], ecological monitoring [5], and precision agriculture [6].

Hyperspectral target detection (HTD), a critical downstream task of HSI processing, aims to identify and locate objects of interest based on spectral and spatial discrepancies. Unlike typical "box" detection methods used in RGB images, which delineate objects with various rectangular or rotated boxes, HTD distinguishes between background and target information at the pixel or subpixel level. This fine-grained detection and joint spectral–spatial discrimination make HTD a powerful tool in current Earth observation and quantitative remote sensing applications [7], [8]. However, it faces two longstanding challenges.

1) *Limited Prior Knowledge:* The task relies on only a few target spectra (usually one) as prior knowledge, which restricts the ability to obtain sufficient features or patterns for effective background-target discrimination, leading to inaccurate detection and poor generalization.
2) *Spectral Variation:* The same material can exhibit diverse spectra, and different materials can appear with similar spectral profiles, making it difficult to achieve reliable and robust performance based solely on spectral discrepancy.

Over the past decades, various approaches have been developed to address these challenges. Early statistical-based methods [9], [10], [11] assumed certain background distributions to determine target pixels. However, their performance may be limited when applied to real-world datasets due to inconsistencies with prior assumptions. Subsequent representation-based methods [12], [13], [14] transformed the detection task into an optimization problem based on the linear mixing model and various regularizations. Despite their physical interpretability, these models may encounter difficulties due to inaccurate regularization constraints and weak nonlinear representations. Given the significant impact of dictionaries in representation-based methods, some studies

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have explored dictionary construction strategies. For instance, Li et al. [15] selected representative pixels based on superpixel segmentation to construct overcomplete background and target dictionaries for sparse representation. Han et al. [16] estimated the background dictionary using K-singular value decomposition and progressively refined the target atom by incorporating high-confidence suspected target spectra to address spectral variation. To eliminate reliance on background priors, Zhu et al. [17] proposed a joint optimization model that alternates between learning the dictionary and coefficients. While these techniques are effective for background-target discrimination, challenges remain in obtaining high-purity dictionaries due to noise and spectral mixing interference. In addition, determining the optimal number of atoms often requires human expertise.

Recently, deep learning, with its powerful feature extraction and parallel computation capabilities, has demonstrated effectiveness and superiority in HTD. However, the scarcity of labeled data poses a significant challenge. To address this, various deep learning-based approaches have been explored, including transfer learning [18], [19], [20], [21], supervised classification [22], [23], [24], [25], and unsupervised and self-supervised detection methods [26], [27], [28], [29], [30], [31]. Transfer learning methods, such as the convolutional neural network (CNN)-based detector (CNND) [18], the meta-learning Siamese network (MLSN) [19], and transfer learning from high-resolution RGB images for HTD (TLH^{2}TD) [20], leverage knowledge from source domains with abundant labeled data and have demonstrated promising performance. However, potential feature drift may hinder their generalization to the target domain. Supervised classification methods expand the training samples via data augmentation and employ a binary classification model for detection. Examples include HTD-Net [22], the two-stream CNN for target detection (TSCNTD) [23], the Siamese Transformer network for target detection (STTD) [24], and the triplet spectral-wise Transformer-based target detector (TSTTD) [25]. However, these methods may introduce bias and noise, leading to mismatches between training and testing data, which can result in false alarms or missed detections. To reduce reliance on labeled data, unsupervised and self-supervised methods have gained traction. Unsupervised methods typically employ autoencoders for HSI reconstruction to extract latent features or produce images that enhance discrimination for detection. Examples include the distance-constrained stacked sparse autoencoder (DCSSAE) [26], which captures discriminative features, and the unconstrained linear mixture model with deep learning (ULMMDL) [27], which enhances target spectral consistency by denoising. Given the availability of sufficient background pixels in the dataset, the target suppression constraint-based background learning (BLTSC) [28] and the orthogonal subspace-guided variational autoencoder (OS-VAE) [29] first reconstructed the background and then separated the target by comparing it with the original HSI. Recently, self-supervised methods [30], [31], such as contrastive learning, have attracted attention for their ability to mine supervisory information from unlabeled data and distinguish between different instances. For example, the self-supervised spectral-level contrastive learning-based HTD (SCLHTD) method [30] applied contrastive learning to promote semantic consistency between feature vectors of the same pixel while distinguishing them from those of different pixels. These unsupervised and self-supervised methods reduce the need for labeled data and show promising performance. However, further improvements in efficient and robust feature extraction are required for real-world applicability. In addition, a well-designed loss function is crucial for enhancing background-target separation.

The current backbones for HTD are mainly based on CNNs and Transformers. CNN-based methods [19], [22], [23], [30] excel at extracting local features but suffer from a limited receptive field, making it difficult to capture nonlocal correlations. In contrast, Transformer-based methods [24], [25], [32], [33] can capture global contextual dependencies based on the multihead self-attention mechanism. However, the computational complexity of Transformers is quadratic to the number of embedding tokens, leading to substantial computational overload and memory requirements as the sequence grows. Recently, structured state space models (SSMs) [34] have gained attention for modeling long sequences with near-linear complexity. Notably, Mamba [35] further enhances state interaction by introducing a selective mechanism (S6) that focuses on the input-relevant information, allowing it to better capture important dependencies across sequences. With its powerful global modeling capabilities and high computational efficiency, Mamba is a promising backbone for various sequence tasks. In HTD, each hyperspectral pixel is a sequence of high-dimensional spectral bands, making it particularly well-suited for capturing long-range spectral dependencies with Mamba. However, the high similarity between adjacent bands in near-continuous spectra can introduce redundancy when processing all bands directly. In addition, focusing on a single-resolution spectral sequence may fail to capture global dependencies and can be more susceptible to noise, easily resulting in unreliable performance, especially when spectral variation occurs. Given these challenges, a more robust Mamba architecture is required for reliable spectral feature extraction in HTD.

Inspired by the above insights, we propose an efficient HTD method with a pyramid SSM (HTD-Mamba). This method employs spectrally contrastive learning to recognize pixel-wise instances by maximizing the feature similarity between each pixel and its transformed view. Thus, detection can be performed by feeding pairs consisting of each testing pixel and the prior target spectrum into the well-trained model. Specifically, an efficient data augmentation technique is proposed to obtain sufficient training samples and leverage spatial information. This technique encodes all surrounding pixels within a patch into a new spectral view of the center pixel, using spectral similarity as weights. In addition, to explore global band correlations, the input pixels are dynamically divided into continuous group-wise spectral embeddings using a 1-D CNN. The efficient architecture, Mamba, is first introduced to HTD to model long-range dependencies of the spectral sequence with linear complexity. Furthermore, to enhance robust feature representations, a pyramid SSM is developed as a backbone

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to extract multiresolution spectral-wise intrinsic features by capturing the global correlation of spectral sequences at different resolutions and fusing features from different levels. Therefore, HTD-Mamba effectively addresses the challenges of limited prior knowledge by constructing sufficient view pairs and mitigates spectral variation by extracting multiresolution discriminative intrinsic features using the pyramid SSM. Experimental results across multiple datasets demonstrate the effectiveness and superiority of HTD-Mamba compared to existing state-of-the-art methods.

In summary, the main contributions can be outlined as follows.

1) We propose a self-supervised spectrally contrastive learning method based on Mamba to address the issues of limited prior knowledge and spectral variation, which recognize pixel-wise instances by maximizing the feature similarity between each pixel and its transformed view. To the best of our knowledge, we are the first to introduce Mamba to HTD, capturing long-range dependencies of the spectral sequence with linear complexity.
2) We propose an efficient data augmentation technique to create sufficient view pairs and leverage spatial contextual information, which encodes all surrounding pixels within a patch into a transformed view of the center pixel.
3) We propose a pyramid SSM to alleviate spectral variation and enhance robust representation, which extracts multiresolution spectral-wise intrinsic features by capturing the global correlation of spectral sequences at different resolutions and fusing features from different levels.

The rest of the article is organized as follows. Section II introduces the preliminary concepts behind Mamba and details our proposed approach. Section III presents the experimental results and analysis. Finally, Section IV concludes the article.

II. Methodology

In this section, we first introduce the basic concepts of Mamba, including SSM, discretization, and the selective scan mechanism. Subsequently, we provide a detailed explanation of the proposed method, covering the overall architecture, module design, and loss function.

A. Preliminary

1) SSM: The SSM is known as a linear time-invariant system that maps a 1-D function or sequence $x(t) \in \mathbb{R}$ to its response $y(t) \in \mathbb{R}$. Mathematically, this process can be formulated using the following linear ordinary differential equations (ODEs):

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) \end{aligned} \tag{1}$$

where $h(t) \in \mathbb{R}^D$ denotes the hidden state with size D, $h'(t)$ denotes the time derivative of $h(t)$, $\mathbf{A} \in \mathbb{R}^{D\times D}$ is the state transition matrix derived from the high-order polynomial projection operator to retain historical information and capture long-range dependencies, and $\mathbf{B} \in \mathbb{R}^{D\times 1}$ and $\mathbf{C} \in \mathbb{R}^{1\times D}$ are the projection matrices controlling the input and output of the system, respectively.

2) Discretization: To integrate the continuous SSM into deep learning algorithms, the discretization process is necessary. Specifically, the ODEs (1) can be discretized using a zero-order hold rule as follows:

$$\begin{aligned} h_t &= \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t \\ y_t &= \mathbf{C}h_t \end{aligned} \tag{2}$$

where

$$\begin{aligned} \bar{\mathbf{A}} &= \exp(\mathbf{\Delta A}) \\ \bar{\mathbf{B}} &= (\mathbf{\Delta A})^{-1}(\exp(\mathbf{\Delta A} - \mathbf{I})) \cdot \mathbf{\Delta B} \end{aligned} \tag{3}$$

and $\mathbf{\Delta}$ is a timescale parameter that balances the influence of the current input and the preceding state. In practice, $\bar{\mathbf{B}}$ is approximated as $\mathbf{\Delta B}$, following the first-order Taylor series.

3) Selective Scan Mechanism: Due to the time-invariant property, i.e., $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{\Delta}$ are independent of the input sequence, the output can be calculated either through recurrence or by global convolution. However, this approach limits context awareness. To address this limitation, a selective scan mechanism is incorporated, where the projection matrices and time scale are derived from the input, ensuring awareness of the contextual information

$$\begin{aligned} S_{\mathbf{B}}(\mathbf{z}) &= \text{Linear}_D(\mathbf{z}) \\ S_{\mathbf{C}}(\mathbf{z}) &= \text{Linear}_D(\mathbf{z}) \\ S_{\mathbf{\Delta}}(\mathbf{z}) &= \text{Broadcast}_N(\text{Linear}_1(\mathbf{z})) \end{aligned} \tag{4}$$

where $\mathbf{z} \in \mathbb{R}^{P\times L\times N}$ is the input sequence, with P as batch size, L as sequence length, and N as embedding size. Specifically, $\text{Linear}_q(\mathbf{z}) = \mathbf{z} \cdot \mathbf{V}_q^{\mathrm{T}}$ represents a linear projection, where $\mathbf{V}_q \in \mathbb{R}^{q\times N}$ is the learnable weight matrix, and $\cdot$ denotes matrix multiplication, transforming $\mathbf{z}$ from $\mathbb{R}^{P\times L\times N}$ to $\mathbb{R}^{P\times L\times q}$. $\text{Broadcast}_q(\mathbf{u}) = \mathbf{u} \cdot \mathbf{E}_q^{\mathrm{T}}$ expands $\mathbf{u} \in \mathbb{R}^{P\times L\times 1}$ along the last dimension to size q, where $\mathbf{E}_q \in \mathbb{R}^{q\times 1}$ is a vector of ones (or alternatively a learnable vector, as described in [35]), resulting in a tensor of shape $\mathbb{R}^{P\times L\times q}$. Thus, the system becomes time-varying, and only recurrence can be adopted. In addition, with the incorporation of parallel scanning and integration with the gated multilayer perceptron (MLP), the SSM evolves into the popular Mamba model, which excels at capturing long-range dependencies with near-linear time complexity and hardware acceleration.

B. Overall Architecture

To address the issues of limited training samples and spectral variation, we propose a self-supervised spectrally contrastive learning method based on Mamba with a pyramid SSM. This method aims to identify pixel-wise instances by maximizing the similarity and consistency between each pixel and its transformed view. Thus, the testing pixel can be determined by pairing it with the prior target spectrum and computing the feature similarity. Fig. 1 illustrates the overall architecture of the proposed method.

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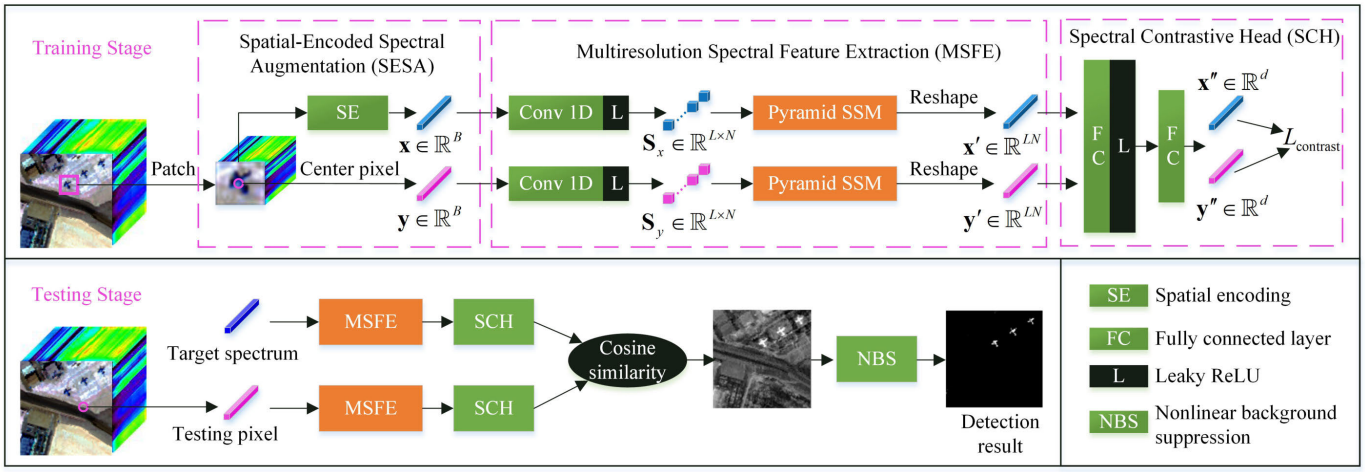


Fig. 1. Overall architecture of the proposed HTD-Mamba.

The training stage encompasses spatial-encoded spectral augmentation (SESA), multiresolution spectral feature extraction (MSFE), and spectral contrastive head (SCH). The SESA block initially creates a new spectral view of the center pixel by encoding contextual information within a patch, thereby generating sufficient view pairs for contrastive learning. Subsequently, the MSFE block divides the pixel into continuous group-wise spectral embeddings, treating them as a spectral sequence. To achieve discriminative intrinsic feature representation, we propose a pyramid SSM that captures multiresolution long-range dependencies within sequences. Finally, the paired extracted features pass through the SCH block to compute similarity, and contrastive loss is applied to ensure model discrimination by pulling positive pairs closer while pushing negative pairs farther apart.

In the testing stage, each testing pixel is paired with the prior target spectrum and then passes through the MSFE and SCH blocks to obtain the discriminative intrinsic feature vectors. The detection result of the ith pixel is obtained by calculating the cosine similarity

$$\mu_i = \frac{f(\mathbf{X}_i)^{\mathrm{T}} f(\mathbf{d})}{\|f(\mathbf{X}_i)\|_2 \cdot \|f(\mathbf{d})\|_2} \tag{5}$$

where $\|\cdot\|_2$ denotes the L2 norm, $\mathbf{X} \in \mathbb{R}^{n \times B}$ denotes the image matrix with n pixels and B bands, $\mathbf{d} \in \mathbb{R}^B$ denotes the target spectrum, and f denotes the backbone consisting of MSFE and SCH. Finally, a nonlinear background suppression (NBS) function is applied to suppress the background

$$\mu_i' = \exp\left(-\frac{(\mu_i - 1)^2}{\delta}\right) \tag{6}$$

where $\delta > 0$ is a tunable parameter that controls the degree of background suppression.

C. Spatial-Encoded Spectral Augmentation

The performance of contrastive learning heavily depends on data augmentation. Current research on spectral data augmentation can be categorized into nonparametric and parametric methods. Nonparametric methods involve direct modifications of the spectra without relying on predefined models, whereas parametric methods are based on specific models or distributions to perform transformations. Nonparametric techniques typically include adding noise [36], band perturbation [37], band masking [38], and smoothing across bands [39]. These methods are computationally efficient due to their simplicity and direct manipulation of the data. For instance, Song et al. [37] employed random spectral shifts to perturb the input patch, while Wang et al. [39] applied Gaussian blurring by convolving each band to create a smoothed version of the original HSI. In contrast, parametric methods, such as random linear mixing [24], [25] and deep generative models [22], [30], [40], offer more powerful modeling capabilities and can generate diverse samples. For example, Jiao et al. [25] generated pseudo-target samples by randomly mixing original pixels with the target spectrum. Wang et al. [30] divided the spectrum into even and odd pairs, which were then processed by two adversarial convolutional autoencoders to generate spectral feature views. In this article, we propose a simple yet efficient spectral augmentation technique that encodes the spatial context within a patch into a new spectral view of the center pixel. This technique aims to leverage local spatial information and provide sufficient view pairs for contrastive learning.

Given a random patch $\mathbf{P} \in \mathbb{R}^{p^2 \times B}$, where the patch size is $p \times p$, and the center pixel is $\mathbf{y} \in \mathbb{R}^B$. By encoding all pixels within the patch using spectral similarity, the new spectral view $\mathbf{x}$ can be expressed as

$$\mathbf{x} = \mathbf{P}^{\mathrm{T}}\mathbf{W} \tag{7}$$

where $\mathbf{W} \in \mathbb{R}^{p^2 \times 1}$ is the weight matrix measuring the similarity between the center pixel and each pixel within the patch. The similarity weight corresponding to the ith pixel within the patch is defined as

$$\mathbf{W}_i = \frac{\exp(c(\mathbf{y}, \mathbf{P}_i))}{\sum_{j=1}^{p^2} \exp(c(\mathbf{y}, \mathbf{P}_j))} \tag{8}$$

where $c(\mathbf{a}, \mathbf{b}) = (\mathbf{a}^{\mathrm{T}}\mathbf{b})/(\|\mathbf{a}\|_2 \cdot \|\mathbf{b}\|_2)$ denotes the cosine similarity function. This mechanism assigns higher weights to

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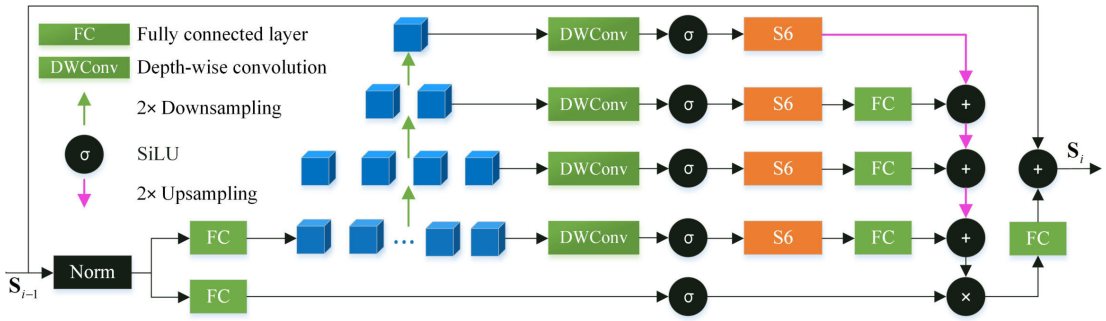


Fig. 2. Flowchart of pyramid SSM.

pixels spectrally similar to the center pixel, ensuring that the transformed view preserves the semantic identity of the center pixel while incorporating local spatial context. As a result, $\mathbf{x}$ is the transformed view of $\mathbf{y}$, preserving the same instance. The pair $(\mathbf{x}, \mathbf{y})$ thus forms a positive sample pair. For the entire image, each pixel is paired with its transformed view from the same spatial location, forming positive sample pairs. In contrast, pairs $(\mathbf{x}, \mathbf{y})$ from different spatial locations represent different pixel instances and thus form negative sample pairs. This augmentation strategy effectively enriches the training data for contrastive learning.

D. Multiresolution Spectral Feature Extraction

To mine discriminative intrinsic features from different spectral views, we treat the group-wise spectral embeddings as a sequence and propose a pyramid SSM to capture multiresolution long-range spectral dependencies.

1) Group-Wise Spectral Embedding: The adjacent bands of the hyperspectral spectrum exhibit high spectral similarity and contain redundant information, which may diminish the effectiveness of spectral feature extraction. Therefore, we adopt a learnable embedding module to convert the input spectrum into group-wise tokens.

Specifically, we use 1-D convolution with a stride to extract local features and divide the spectrum into group representations

$$\mathbf{S}_x = \sigma_1(\mathrm{Conv}(\mathbf{x}, m, s, N)) \tag{9}$$

where Conv denotes the 1-D convolution, m denotes the kernel size, $s = \max(1, \lfloor m/4 \rfloor)$ denotes the stride, $\lfloor \cdot \rfloor$ denotes the rounding down operation, N denotes the number of kernels, and σ_1 is the leaky rectified linear unit (leaky ReLU) [41] function for nonlinear representation. After this process, paired spectral samples $(\mathbf{x}, \mathbf{y})$ can be converted to group-wise embeddings $(\mathbf{S}_x \in \mathbb{R}^{L\times N}, \mathbf{S}_y \in \mathbb{R}^{L\times N})$, where L denotes the number of spectral groups, satisfying $L = \lfloor (B-m)/s \rfloor + 1$. Thus, numerous spectral tokens can be obtained to extract discriminative intrinsic features.

2) Pyramid SSM: The resulting group-wise spectral embeddings can be viewed as a spectral sequence. Considering the linear complexity of SSM in capturing long-range dependencies and the benefits of multiresolution analysis, we propose a pyramid SSM to extract multiresolution spectral-wise intrinsic features.

The flowchart of the proposed pyramid SSM is shown in Fig. 2. It comprises bottom-up downsampling, global spectral feature extraction, top-down upsampling, and feature fusion, and the gated block. In each deep layer, the input sequence first undergoes a normalization operation to reduce the internal covariate shift and stabilize the training process. In this work, the root mean square layer normalization (RMSNorm) [42] is applied to improve the computational efficiency by eliminating the mean calculation required in the layer normalization

$$\bar{\mathbf{S}}_{i-1}^{j} = \frac{\mathbf{S}_{i-1}^{j}}{\mathrm{rms}(\mathbf{S}_{i-1})}\mathbf{g}_{i-1}^{j} \tag{10}$$

where rms denotes the root mean square function, $\mathbf{S}_{i-1}^{j}$ denotes the jth channel of the $(i-1)$th layer input sequence, and $\mathbf{g}$ denotes the learnable weight. Then, the normalized sequence is projected into two representations for feature extraction and gated operation, each using a fully connected (FC) layer

$$\begin{aligned} \mathbf{Z}_1 &= \mathrm{Linear}_{2N}(\bar{\mathbf{S}}_{i-1}) \\ \mathbf{Z}_2 &= \mathrm{Linear}_{2N}(\bar{\mathbf{S}}_{i-1}). \end{aligned} \tag{11}$$

In the bottom-up downsampling process, the spectral sequence is gradually downsampled to obtain multiresolution representations, which facilitates comprehensive analysis. Specifically, the adjacent hierarchical sampling is performed using convolution with stride

$$\mathbf{Z}_1^k = \mathrm{Conv}\left(\mathbf{Z}_1^{k-1}, 3, 2, 2^{k+1}N\right) \tag{12}$$

where $\mathbf{Z}_1^0 = \mathbf{Z}_1$, $k \in \{1, 2, 3\}$ denotes the kth pyramid level. The kernel size is fixed at 3, the stride is 2 to perform 2× downsampling, and the token size is doubled to maintain information capacity.

For each pyramid level, the spectral sequence first passes through a 1-D depth-wise convolution with activation to extract local features and then undergoes an S6 module to capture long-range dependencies among tokens. This process can be expressed as

$$\bar{\mathbf{Z}}_1^k = \mathrm{S6}\left(\sigma_2\left(\mathrm{DWConv}\left(\mathbf{Z}_1^k, 3, 2^{k+1}N\right)\right)\right) \tag{13}$$

where $k \in \{0, 1, 2, 3\}$, DWConv denotes the 1-D depth-wise convolution, the kernel size is 3, the number of groups is $2^{k+1}N$, σ_2 denotes the sigmoid-weighted linear unit (SiLU) [43] function, and padding is used in the convolution to keep the size fixed.

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Algorithm 1 S6

Require: Spectral sequence $\mathbf{z}$: (P, L, N)
Ensure: Spectral sequence $\bar{\mathbf{z}}$: (P, L, N)
$\mathbf{B}$: $(P, L, D) \leftarrow S_{\mathbf{B}}(\mathbf{z})$
$\mathbf{C}$: $(P, L, D) \leftarrow S_{\mathbf{C}}(\mathbf{z})$
$\mathbf{\Delta}$: $(P, L, N) \leftarrow \log(1 + \exp(S_{\mathbf{\Delta}}(\mathbf{z}) + \text{Parameter}^{\mathbf{\Delta}}))$
$\bar{\mathbf{A}}$: $(P, L, N, D) \leftarrow \mathbf{\Delta} \otimes \text{Parameter}^{\mathbf{A}}$
$\bar{\mathbf{B}}$: $(P, L, N, D) \leftarrow \mathbf{\Delta} \otimes \mathbf{B}$
$\bar{\mathbf{z}}$: $(P, L, N) \leftarrow \text{SSM}(\bar{\mathbf{A}}, \bar{\mathbf{B}}, \mathbf{C})(\mathbf{z})$
Return : $\bar{\mathbf{z}}$

In the top-down upsampling and feature fusion process, low-level features are recovered from the adjacent high level and then fused with the extracted global features from the same level. This process can be expressed as

$$\begin{aligned} \hat{\mathbf{Z}}_1^{k-1} &= \text{DConv}\left(\hat{\mathbf{Z}}_1^{k}, 3, 2, 2^k N\right)) \\ \hat{\mathbf{Z}}_1^{k-1} &= \hat{\mathbf{Z}}_1^{k-1} + \text{Linear}_{2^k N}\left(\bar{\mathbf{Z}}_1^{k-1}\right) \end{aligned} \tag{14}$$

where $k \in \{3, 2, 1\}$, $\hat{\mathbf{Z}}_1^3 = \bar{\mathbf{Z}}_1^3$, DConv denotes the transposed 1-D convolution with a kernel size of 3 and a stride of 2 to perform $2\times$ upsampling. The token size is halved to maintain information capacity. The FC layer is used to perform flexible linear transformation in preparation for fusion.

The gated block is adopted to focus on useful information while filtering out redundant information. This process can be expressed as

$$\mathbf{Z} = \hat{\mathbf{Z}}_1^0 \odot \sigma_2(\mathbf{Z}_2) \tag{15}$$

where $\odot$ denotes element-wise multiplication. Finally, an FC layer is used to adjust the token size, and a residual connection is applied to reduce the loss of information

$$\mathbf{S}_i = \mathbf{S}_{i-1} + \text{Linear}_N(\mathbf{Z}). \tag{16}$$

An overview of the proposed pyramid SSM is summarized in Algorithms 1 and 2, where P denotes the batch size, $\otimes$ denotes the outer product, and $\text{Parameter}^{\mathbf{o}}$ denotes the learnable parameter.

E. Spectral Contrastive Head

The SCH further projects the extracted multiresolution spectral features into the feature space using two FC layers, which can be expressed as

$$\mathbf{x}'' = \text{Linear}_d(\sigma_1(\text{Linear}_{2d}(\mathbf{x}'))) \tag{17}$$

where $\mathbf{x}' \in \mathbb{R}^{LN}$ denotes the reshaped spectral features, and d denotes the size of feature vectors.

To maximize the similarity of positive sample pairs while minimizing the similarity of negative sample pairs, contrastive learning is required. In this article, the feature vector pairs $(\mathbf{x}'', \mathbf{y}'')$ from the same location construct the positive sample pairs, while those from different locations construct the negative ones. For example, given a batch of $(\mathbf{x}'', \mathbf{y}'')$, for a specific sample $\mathbf{x}''^{(i)}$, there are P pairs in total, where $(\mathbf{x}''^{(i)}, \mathbf{y}''^{(i)})$ is the only positive pair while the others are negative pairs. Thus, the loss for $\mathbf{x}''^{(i)}$ can be expressed as

$$\ell(\mathbf{x}''^{(i)}, \mathbf{y}''^{(i)}) = -\log \frac{\exp\left(c\left(\mathbf{x}''^{(i)}, \mathbf{y}''^{(i)}\right)\Big/\alpha\right)}{\sum_{j=1}^{P} \exp\left(c\left(\mathbf{x}''^{(i)}, \mathbf{y}''^{(j)}\right)/\alpha\right)} \tag{18}$$

where α is the temperature parameter to control the degree of attention to negative pairs. Taking into account learning the similarity of all the positive pairs within a batch, the spectral contrastive loss can be expressed as

$$L_{\text{contrast}} = \frac{1}{P} \sum_{i=1}^{P} \ell\left(\mathbf{x}''^{(i)}, \mathbf{y}''^{(i)}\right). \tag{19}$$

Through spectrally contrastive learning, semantic-similar samples are brought closer together while dissimilar samples are pushed further apart, enabling the model to distinguish the spectral similarities and dissimilarities of different instances.

Algorithm 2 Pyramid SSM

Require: Spectral sequence $\mathbf{S}_{i-1}$: (P, L, N)
Ensure: Spectral sequence $\mathbf{S}_i$: (P, L, N)
/* Normalization and projection */
$\bar{\mathbf{S}}_{i-1}$: $(P, L, N) \leftarrow \text{RMSNorm}(\mathbf{S}_{i-1})$
$\mathbf{Z}_1$: $(P, L, 2N) \leftarrow \text{Linear}_{2N}(\bar{\mathbf{S}}_{i-1})$
$\mathbf{Z}_2$: $(P, L, 2N) \leftarrow \text{Linear}_{2N}(\bar{\mathbf{S}}_{i-1})$
/* Downsampling and feature extraction */
$L_0 = L$, $\mathbf{Z}_1^0 = \mathbf{Z}_1$
$\bar{\mathbf{Z}}_1^0$: $(P, L_0, 2N) \leftarrow \text{S6}(\sigma_2(\text{DWConv}(\mathbf{Z}_1^0)))$
for $k = 1$ to 3 **do**
  $L_k = L_{k-1}//2$
  $\mathbf{Z}_1^k$: $(P, L_k, 2^{k+1}N) \leftarrow \text{Conv}(\mathbf{Z}_1^{k-1})$
  $\bar{\mathbf{Z}}_1^k$: $(P, L_k, 2^{k+1}N) \leftarrow \text{S6}(\sigma_2(\text{DWConv}(\mathbf{Z}_1^k)))$
end for
/* Upsampling and feature fusion */
$\hat{\mathbf{Z}}_1^3 = \bar{\mathbf{Z}}_1^3$
for $k = 3$ to 1 **do**
  $\hat{\mathbf{Z}}_1^{k-1}$: $(P, L_{k-1}, 2^k N) \leftarrow \text{DConv}(\hat{\mathbf{Z}}_1^k)$
  $\hat{\mathbf{Z}}_1^{k-1}$: $(P, L_{k-1}, 2^k N) \leftarrow \hat{\mathbf{Z}}_1^{k-1} + \text{Linear}_{2^k N}(\bar{\mathbf{Z}}_1^{k-1})$
end for
/* Gated block */
$\mathbf{Z}$: $(P, L, 2N) \leftarrow \hat{\mathbf{Z}}_1^0 \odot \sigma_2(\mathbf{Z}_2)$
/* Residual connection */
$\mathbf{S}_i$: $(P, L, N) \leftarrow \mathbf{S}_{i-1} + \text{Linear}_N(\mathbf{Z})$
Return : $\mathbf{S}_i$

III. Experimental Results and Analysis

In this section, we compare the proposed HTD-Mamba method with state-of-the-art methods on four benchmark datasets to verify its effectiveness.

A. Datasets and Experimental Setup

1) Datasets: The first two datasets, San Diego I and II, were captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over San Diego Airport, San Diego, CA, USA. Both HSIs have a size of $100 \times 100 \times 224$ with a spatial resolution of 3.5 m, covering a wavelength range

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from 400 to 2500 nm. After removing low-quality bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224), 189 bands are preserved for experiments. The targets to be detected are three airplanes, consisting of 58 pixels in San Diego I and 134 pixels in San Diego II, respectively.

The third dataset, Los Angeles, was collected by AVIRIS over an airport scene in Los Angeles, California, USA. The HSI has a size of $100 \times 100 \times 205$, with 100×100 pixels and 205 spectral bands, covering a wavelength range from 400 to 2500 nm. The spatial resolution of this dataset is 7.1 m. The targets to be detected are two airplanes, consisting of 87 pixels.

The fourth dataset, Pavia, was captured by the Reflective Optics System Imaging Spectrometer (ROSIS-03) sensor over a beach area in Pavia, Italy. The HSI has a spatial size of 150×150 with a spatial resolution of 1.3 m. It consists of 115 spectral bands ranging from 430 to 860 nm. After removing low-quality bands, 102 bands remain for experiments. Some man-made objects are labeled as targets, consisting of 68 pixels. Both the Los Angeles and Pavia datasets are available online.[1]

In the experiments, the pixel closest to the averaged spectrum of all the target pixels is selected as the target spectrum, as it effectively represents the fundamental spectral characteristics of the object to be detected. In practice, one can also select the target spectrum from a spectral library if available, or by selecting a valid random target pixel from the sampling area, or by calculating the average spectrum of the target area. The false-color images and the ground-truth maps are displayed in Fig. 3.

2) Quality Metrics: To assess the quality of the detection map, both qualitative and quantitative metrics are adopted. Specifically, the visualized detection map, receiver operating characteristic (ROC) curve, and background-target separability diagram are utilized for qualitative assessment. The ROC curve is a function of false alarm rate P_f and detection probability P_d under different thresholds τ. For a comprehensive analysis, 3-D ROC curves [44] are adopted, where three unfolded 2-D ROC curves of (P_f, P_d), (P_d, τ), and (P_f, τ) are obtained to measure detection effectiveness, target detectability, and background suppression, respectively. The background-target separability diagram is plotted using the Box–Whisker function to show the distinction between the background and target. For quantitative evaluation, three areas under the ROC curves (AUCs) are applied, i.e., $\mathrm{AUC}_{(P_f,P_d)}$, $\mathrm{AUC}_{(\tau,P_d)}$, and $\mathrm{AUC}_{(\tau,P_f)}$. In addition, two composite metrics are also calculated to evaluate overall quality

$$\mathrm{AUC}_{\mathrm{OA}} = \mathrm{AUC}_{(P_f,P_d)} + \mathrm{AUC}_{(\tau,P_d)} - \mathrm{AUC}_{(\tau,P_f)} \tag{20}$$

$$\mathrm{AUC}_{\mathrm{SNPR}} = \mathrm{AUC}_{(\tau,P_d)} / \mathrm{AUC}_{(\tau,P_f)}. \tag{21}$$

The detailed properties of these five metrics are listed in Table I.

3) Implementation Details: The proposed HTD-Mamba is implemented using Python 3.10.4 and PyTorch 2.0.1, with an Intel Core i9-10900X CPU, 32 GB of RAM, and an NVIDIA TITAN RTX with 24 GB of GPU memory. The batch size P is set to 80. For each batch, the patch size surrounding the center pixel is set to 11×11, 13×13, 11×11, and 5×5 for San Diego I, San Diego II, Los Angeles, and Pavia, respectively. In the group-wise spectral embedding, the group length m is set to 30, 5, 5, and 15 for the four datasets, respectively. The embedding size N is set to 16 for all four datasets. In the feature extraction and projection blocks, the depth of the pyramid SSM is set to 1, the hidden state size D is set to 16, and the dimension of the final feature vectors d is set to 32 for all four datasets. For the contrastive loss, the temperature parameter α is fixed at 0.1 for all four datasets. The AdamW optimizer is selected for training, where the weight-decay parameter is set to 10^{-4}. The initial learning rate is set to 10^{-4}, which increases via a linear warmup scheduler in the first 10% epochs and then decreases via a cosine scheduler. The total number of epochs is 200. In the background suppression process, the parameter δ is set to 0.1 for all four datasets.

TABLE I
PROPERTIES OF QUANTITATIVE METRICS

Property	$\mathrm{AUC}_{(P_f,P_d)}$	$\mathrm{AUC}_{(\tau,P_d)}$	$\mathrm{AUC}_{(\tau,P_f)}$	$\mathrm{AUC}_{\mathrm{OA}}$	$\mathrm{AUC}_{\mathrm{SNPR}}$
Focus	Effectiveness	Detectability	False alarm	Overall	Overall
Range	[0, 1]	[0, 1]	[0, 1]	[-1, 2]	[0, $+\infty$)
Trend	↑	↑	↓	↑	↑
Optimum	1	1	0	2	$+\infty$

B. Parameter Analysis

In this section, we discuss the effects of several key hyperparameters, namely, the patch size p, the spectral group length m, the embedding size N, and the network depth l.

1) Patch Size p: In the SESA process, the transformed view of the center pixel is affected by the patch size. To explore its impact on detection quality, we conducted experiments by varying p within the range of [3, 19], with p being an odd number. The experimental results for the four datasets are shown in Fig. 4(a). It can be observed that the sensitivity to p varies across different datasets. The performance is relatively stable for San Diego I, shows some fluctuations for San Diego II and Los Angeles, and exhibits the greatest variability for Pavia. This variation is related to the size and distribution of the targets in the different scenes. Considering both detection performance and computational cost, the selected p values for the four datasets are 11, 13, 11, and 5, respectively.

2) Spectral Group Length m: The spectral group length m determines the length of the spectral sequence. Smaller m values result in more groups and longer token sequences, while larger m values yield shorter token sequences. The sequence length can significantly impact spectral feature extraction, thereby affecting detection quality. To select an appropriate group length, we varied m within the range of [5, 40], with a step size of 5. The experimental results for the four datasets are shown in Fig. 4(b). As observed from the figure, the AUC values fluctuate within a relatively small range for the San Diego I, San Diego II, and Los Angeles datasets as m varies. However, the Pavia dataset exhibits significant fluctuations,

[1] http://xudongkang.weebly.com/

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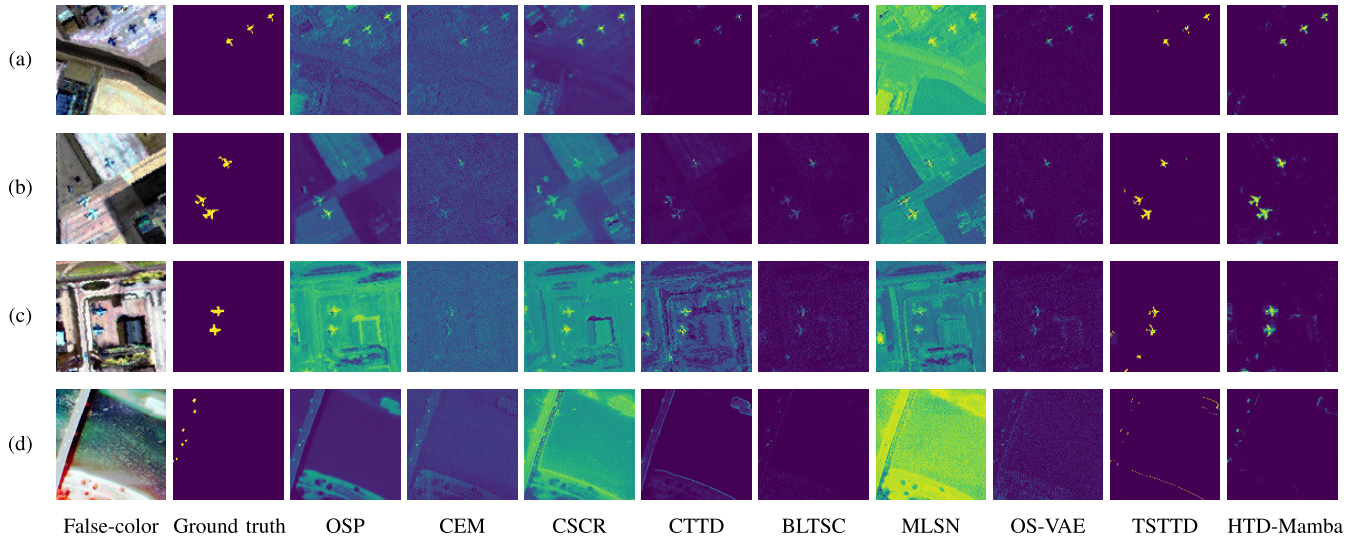


Fig. 3. Visualized results of the competing methods on (a) San Diego I, (b) San Diego II, (c) Los Angeles, and (d) Pavia.

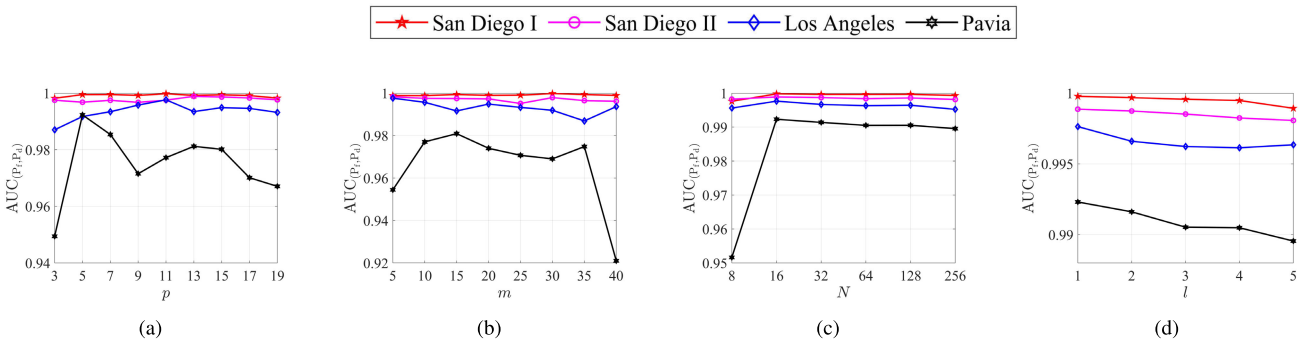


Fig. 4. Parameter effect analysis of (a) patch size p, (b) spectral group length m, (c) embedding size N, and (d) network depth l.

likely due to large changes in local spectral characteristics. Considering the diversity of the formed spectral sequences and computational cost, the selected m values for the four datasets are 30, 5, 5, and 15, respectively.

3) Embedding Size N: The embedding size N represents the dimensionality of the spectral tokens, which also influences detection results. To select an appropriate N value, we varied it within the range of [8, 256] in powers of 2. The experimental results are shown in Fig. 4(c). As observed, for all four datasets, detection accuracy initially increases with increasing N, but then stabilizes or decreases. This is because increasing the feature dimensionality within a certain range enhances the feature representation capability, but excessively large feature dimensions can lead to overfitting. To balance detection accuracy and computational complexity, we set N as 16 for all four datasets.

4) Network Depth l: The network depth l represents the number of deep layers in the pyramid SSM. To investigate its impact on detection performance, we compared the AUC scores for network depths ranging from 1 to 5. The experimental results are shown in Fig. 4(d). As shown in the figure, detection accuracy remains stable or slightly decreases with increasing l. This is because the pyramid SSM inherently possesses the ability to extract global spectral features with multiresolution using shallow layers. When the number of layers increases, the model may overfit, leading to decreased accuracy and increased computational complexity. Therefore, we set l to 1 for all datasets to ensure a lightweight model while maintaining detection effectiveness.

C. Performance Comparison

In this work, we select eight representative advanced methods for comparison: orthogonal subspace projection (OSP) [10], constrained energy minimization (CEM) [45], combined sparse and collaborative representation (CSCR)[2] [13], chessboard-shaped topology for HTD (CTTD)[3] [46], BLTSC[4] [28], MLSN[5] [19], OS-VAE[6] [29], and TSTTD[7] [25]. Among these, CEM and OSP are statistical-based methods, CSCR is a representation-based method, CTTD is a topological method, and BLTSC, MLSN, OS-VAE, and TSTTD are deep learning-based methods.

For a fair comparison, the key parameters of the competing methods are tuned to optimal levels. Specifically, for the

[2]https://fdss.bit.edu.cn/pub/fsyxhyxtktz/yjdw/js/b153191.htm
[3]https://github.com/sxt1996/CTTD
[4]https://github.com/zhangxin-xd/BLTSC
[5]https://github.com/YuleiWang1/MLSN
[6]https://github.com/CX-He/OS-VAE
[7]https://github.com/shendb2022/TSTTD

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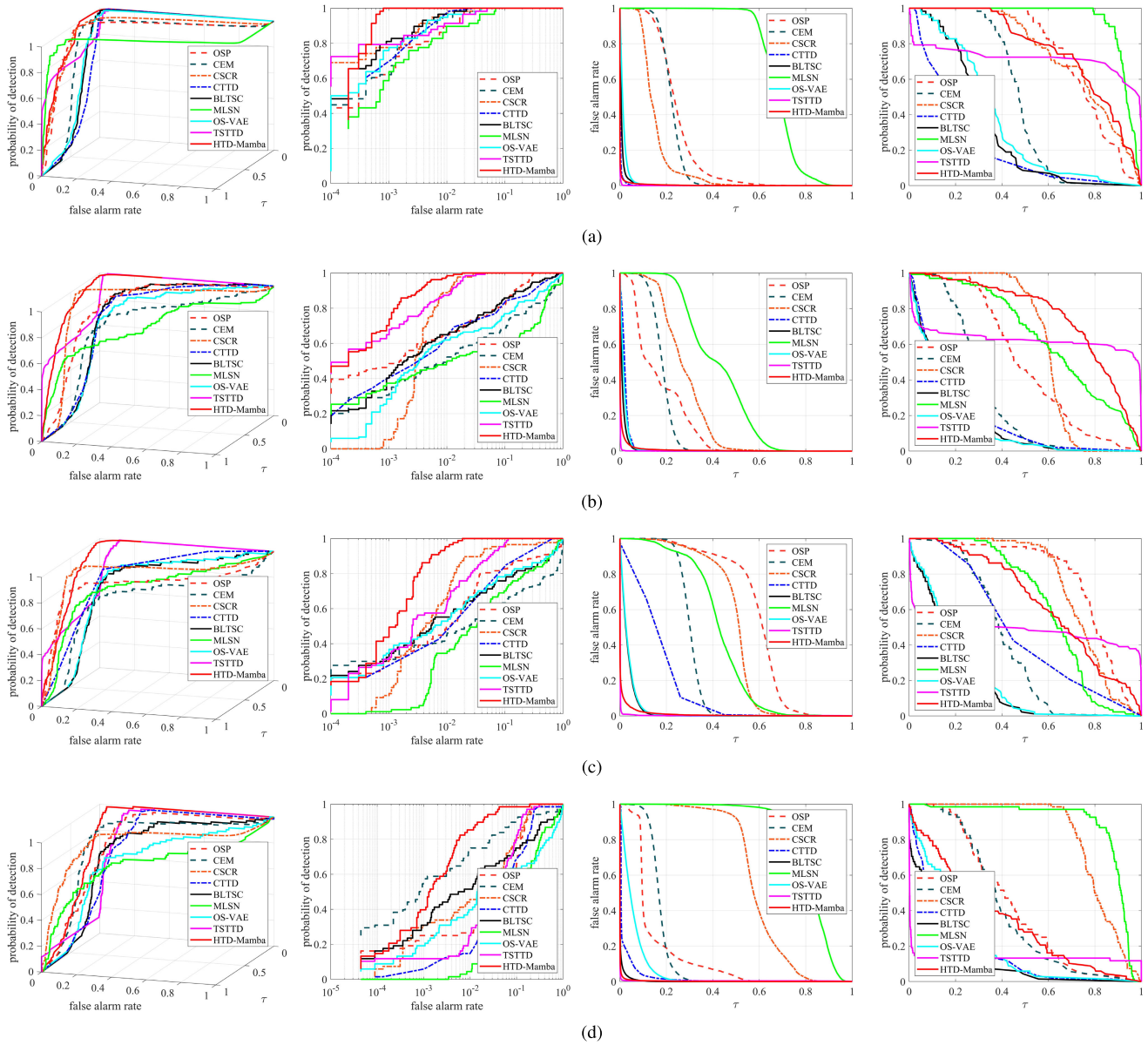


Fig. 5. ROC curves of the competing methods on (a) San Diego I, (b) San Diego II, (c) Los Angeles, and (d) Pavia. From (left to right) 3-D ROC curve, 2-D ROC curve of (P_f, P_d), 2-D ROC curve of (τ, P_f), and 2-D ROC curve of (τ, P_d).

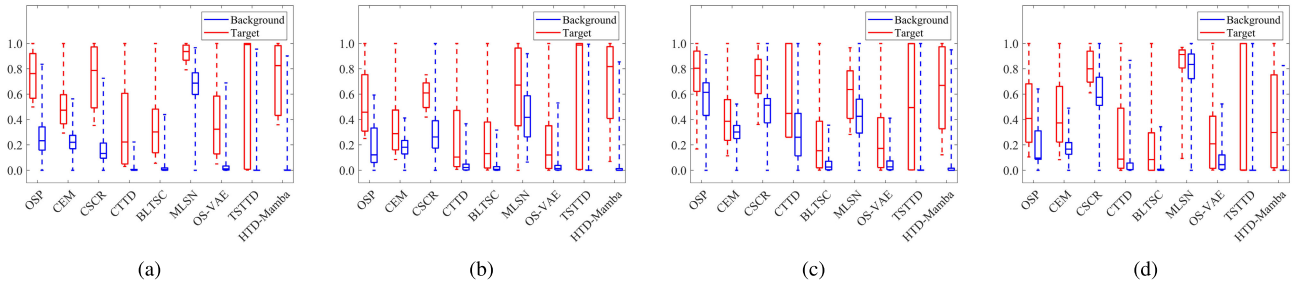


Fig. 6. Background-target separation diagrams of competing methods on four datasets. (a) San Diego I. (b) San Diego II. (c) Los Angeles. (d) Pavia.

OSP detector, the automatic target generation process (ATGP) [47] is applied to obtain the background subspace, where the number of bases is selected as 22, 2, 1, and 1 for San Diego I, San Diego II, Los Angeles, and Pavia, respectively. In CSCR, the dual window size $(w_{\text{out}}, w_{\text{in}})$ is set to $(13, 11)$, $(15, 3)$, $(15, 13)$, and $(15, 13)$ for the four datasets, respectively. In addition, the two regularization parameters (λ, β) are set to $(0.1, 1)$ for all datasets. For CTTD, the vertical division for

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TABLE II
AUC SCORES AND RUNNING TIME (IN SECONDS) OF COMPETING METHODS ON THE FOUR DATASETS. BOLD REPRESENTS THE BEST RESULT WHILE UNDERLINED THE SECOND

Dataset		OSP	CEM	CSCR	CTTD	BLTSC	MLSN	OS-VAE	TSTTD	HTD-Mamba
San Diego I	$AUC_{(P_f,P_d)}$	0.9972	0.9977	0.9985	0.9982	0.9987	0.9940	0.9978	0.9971	**0.9998**
	$AUC_{(\tau,P_f)}$	0.2461	0.2224	0.1525	0.0055	0.0091	0.6862	0.0163	**0.0007**	0.0036
	$AUC_{(\tau,P_d)}$	0.7616	0.4860	0.7634	0.3461	0.3285	**0.9335**	0.3637	0.7295	0.7817
	AUC_{OA}	1.5127	1.2613	1.6094	1.3388	1.3181	1.2413	1.3452	1.7259	**1.7779**
	AUC_{SNPR}	3.0947	2.1853	5.0059	62.9273	36.0989	1.3604	22.3129	**1042.1429**	217.1389
	Running time	0.4021	**0.3320**	3.6459	0.3480	2.4912	12.6212	2.4184	2.7609	1.6895
San Diego II	$AUC_{(P_f,P_d)}$	0.9599	0.8468	0.9952	0.9220	0.9385	0.7944	0.8875	0.9969	**0.9989**
	$AUC_{(\tau,P_f)}$	0.1672	0.1818	0.2753	0.0286	0.0121	0.4179	0.0183	**0.0018**	0.0079
	$AUC_{(\tau,P_d)}$	0.5156	0.3136	0.6019	0.2263	0.1706	0.6528	0.1623	0.6316	**0.7635**
	AUC_{OA}	1.3083	0.9786	1.3218	1.1197	1.0970	1.0293	1.0315	1.6267	**1.7545**
	AUC_{SNPR}	3.0837	1.7250	2.1863	7.9126	14.0992	1.5621	8.8689	**350.8889**	96.6456
	Running time	**0.0625**	0.4118	10.0797	0.2980	2.4624	12.4940	2.4111	2.4944	1.7474
Los Angeles	$AUC_{(P_f,P_d)}$	0.8937	0.7588	0.9675	0.9180	0.8696	0.8521	0.8773	0.9838	**0.9976**
	$AUC_{(\tau,P_f)}$	0.5893	0.3019	0.4915	0.3192	0.0330	0.4276	0.0349	**0.0043**	0.0102
	$AUC_{(\tau,P_d)}$	**0.7842**	0.4045	0.7452	0.6865	0.2016	0.6290	0.2133	0.4948	0.6552
	AUC_{OA}	1.0886	0.8614	1.2212	1.2853	1.0382	1.0535	1.0557	1.4743	**1.6426**
	AUC_{SNPR}	1.3307	1.3398	1.5162	2.1507	6.1091	1.4710	6.1117	**115.0698**	64.2353
	Running time	**0.0591**	0.3349	4.3352	0.2420	2.3861	12.6853	2.4097	2.7999	1.7729
Pavia	$AUC_{(P_f,P_d)}$	0.9239	0.9419	0.9489	0.9049	0.8738	0.7867	0.7823	0.9468	**0.9923**
	$AUC_{(\tau,P_f)}$	0.1457	0.1704	0.5936	0.0233	0.0038	0.8239	0.0567	0.0042	**0.0025**
	$AUC_{(\tau,P_d)}$	0.4331	0.4089	0.8165	0.1655	0.1359	**0.8848**	0.2181	0.1337	0.3481
	AUC_{OA}	1.2113	1.1804	1.1718	1.0471	1.0059	0.8476	0.9437	1.0763	**1.3379**
	AUC_{SNPR}	2.9725	2.3996	1.3755	7.1030	35.7632	1.0739	3.8466	31.8333	**139.2400**
	Running time	0.0996	**0.0883**	6.7446	0.5490	2.6024	26.3778	1.4024	2.8134	2.7239

spatial disassembly ver and the horizontal division for spectral disassembly hor, are set to (18, 2), (14, 4), (3, 4), and (18, 7) for the four datasets, respectively. For BLTSC, the binarization threshold k in coarse detection is fixed at 0.15, the learning rate is 10^{-4}, the batch size is 64, the number of epochs is 500, and the parameter λ in nonlinear background suppression is set to 10^3, 10^4, 10^4, and 10 for the four datasets, respectively. For MLSN, the batch size, learning rate, and epoch are set to 128, 10^{-3}, and 50, respectively. In the process of guided image filtering, the local window radius is set to 2, and the penalty value is set to 0.04 for all datasets. For OS-VAE, the number of hidden nodes, the network depth, the regularization parameter ρ, and the background suppression parameter α are, respectively, set to 30, 2, 10^{-5}, and 0.1 for all datasets. For TSTTD, the mixture ratio of background in synthesizing target samples is randomly set between 0 and 0.1. During the training process, the batch size is set to 64, the learning rate is 10^{-4}, and the total number of epochs is 20.

For an intuitive comparison, Fig. 3 presents the detection results obtained by all competing methods across four benchmark datasets. It illustrates that the proposed HTD-Mamba outperforms other methods by achieving a promising balance between target highlighting and background suppression. The classical detector CEM exhibits weak performance in accurately identifying targets while suppressing the background. Detectors such as OSP, CSCR, and MLSN can detect most target pixels but suffer from poor background suppression. In contrast, BLTSC, OS-VAE, and TSTTD excel at eliminating background interference at the expense of losing significant target information. The CTTD method performs unsteadily, either missing useful information or producing numerous false alarms. Compared to other methods, HTD-Mamba achieves the most satisfactory results in simultaneously highlighting targets and suppressing the background. The detection maps obtained by HTD-Mamba closely align with the ground truths, effectively preserving the shapes and contours of the objects of interest.

Fig. 5 presents the ROC curves of the competing methods across four datasets, including a 3-D ROC curve and three unfolded 2-D ROC curves. For the ROC (P_f, P_d), the closer the curve is to the upper left corner, the better the performance. It can be observed that the proposed HTD-Mamba leads in all four datasets, demonstrating superior detection effectiveness. For the ROC (τ, P_f), which reflects the quality of background suppression, the nearer the curve is to the lower left corner, the better the performance. Among the competing methods, TSTTD and HTD-Mamba achieve the most satisfactory performance in background suppression. In addition, the closer the curve is to the upper right corner in the ROC (τ, P_d) space, the better the detection ability of the method. Although HTD-Mamba does not outperform CSCR and MLSN on the Pavia dataset, its performance remains competitive. Considering the comprehensive capabilities of target prominence and background suppression, our proposed HTD-Mamba achieves

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the most promising and reliable detection performance across all four datasets.

The ROC analysis alone cannot accurately evaluate detection quality due to overlapping results from multiple methods. Therefore, Table II provides a detailed comparison of the AUC scores obtained by competing methods on the four datasets for quantitative evaluation. The best results are marked in bold, while the second-best are underlined. It is evident that the proposed HTD-Mamba achieves the optimal results for the most important $AUC_{(P_f,P_d)}$ score across the four datasets, which are approximately 0.0011, 0.0020, 0.0138, and 0.0434 higher than those of the second-best, respectively. This indicates its outstanding performance in detection effectiveness. While OSP and MLSN achieve superior $AUC_{(\tau,P_d)}$ results on the Los Angeles and Pavia datasets, they perform relatively poorly in terms of background suppression due to high $AUC_{(\tau,P_f)}$ scores. In contrast, BLTSC and TSTTD obtain satisfactory $AUC_{(\tau,P_f)}$ results across all four datasets but suffer a severe loss of target information. Our proposed HTD-Mamba maintains a promising balance between target detection and background suppression, due to competitive $AUC_{(\tau,P_d)}$ and $AUC_{(\tau,P_f)}$ scores. In addition, HTD-Mamba also achieves superior AUC_{OA} and AUC_{SNPR} results across the four datasets, indicating comprehensive detection performance. Table II also presents the running time of the competing methods. It is evident that OSP, CEM, and CTTD have lower computational costs due to their simpler algorithms. Our HTD-Mamba achieves competitive computational speed among deep detectors due to the hardware parallel scanning technique, facilitating practical application. In summary, the quantitative assessment provided by AUC scores and running time demonstrates the outstanding performance of the proposed HTD-Mamba.

To further compare the background-target separation performance of the competing methods, Fig. 6 shows the separation diagrams for the four datasets. It is evident that our proposed HTD-Mamba achieves the most satisfactory performance, as there is significant distance and minimal overlap between the target and background boxes across all four datasets. Although BLTSC, TSTTD, and HTD-Mamba all achieve superior background suppression performance with small-sized background boxes, our proposed HTD-Mamba demonstrates a more promising target-background separation. Overall, HTD-Mamba consistently produces satisfactory visual detection maps, achieves outstanding quantitative results, and demonstrates promising target-background separation capabilities across all four datasets.

D. Ablation Analysis

In this section, we verify and analyze the effectiveness of the key contributions of the proposed method.

1) Effectiveness of SESA: The proposed SESA technique encodes surrounding contextual information to create a new spectral view of the center pixel, providing sufficient view pairs for contrastive learning. To verify its effectiveness, we compare SESA with the no-transform strategy (referred to as self-similarity) and several spectral data augmentation methods, including Dropout [48], Gaussian noise, variational autoencoder (VAE) [49], and adversarial autoencoder (AAE) [50]. Specifically, self-similarity treats each pixel and its copy as a positive pair, while different pixels are treated as negative pairs. Dropout randomly deactivates a certain percentage (0.5 in this article) of the bands of each pixel to create a new spectral view, which is then paired with the original pixel to construct the positive sample. The Gaussian noise method adds random noise (20 dB in this article) to the pixel to generate a degraded view simulating complex scenes. Both VAE and AAE are autoregressive methods that encode the spectral information into a latent space, either by maximizing the variational lower bound or through adversarial training. Pixels can be paired with their regressed variants generated by VAE or AAE to construct view pairs for training. We adopt the same autoencoder architecture used in [30] for both VAE and AAE, which includes a spectral residual channel attention module and a simple discriminator composed of two FC layers for AAE. For a fair comparison, the testing pixel is directly paired with the target spectrum during inference.

TABLE III
PERFORMANCE COMPARISON OF COMPETING SPECTRAL AUGMENTATION METHODS ON FOUR DATASETS

Method	San Diego I	San Diego II	Los Angeles	Pavia
Self-Similarity	0.9873	0.9425	0.8544	0.8563
Dropout	0.9917	0.9899	0.9668	0.8868
Gaussian Noise	0.9915	0.9758	0.9549	0.8966
VAE	0.9857	0.9656	0.8886	0.9129
AAE	0.9979	0.9880	0.9317	0.8863
SESA	**0.9998**	**0.9989**	**0.9976**	**0.9923**

TABLE IV
EFFECTIVENESS OF PYRAMID SSM ON FOUR DATASETS

Pyramid SSM	San Diego I	San Diego II	Los Angeles	Pavia
Without	0.9986	0.9978	0.9853	0.9127
With (Level=1)	0.9991	0.9979	0.9921	0.9336
With (Level=2)	0.9991	0.9972	0.9938	0.9824
With (Level=3)	0.9995	0.9985	0.9950	0.9859
With (Level=4)	**0.9998**	**0.9989**	**0.9976**	**0.9923**

Table III shows the compared $AUC_{(P_f,P_d)}$ scores on the four datasets. It can be observed that SESA outperforms other competing data augmentation methods. The self-similarity method cannot effectively recognize similar instances because the exact match criterion is too strict. Both AAE and VAE are parametric and require pretraining. Moreover, these competing methods do not consider spatial contextual information. In contrast, the proposed SESA not only makes full use of spatial information but is also computationally efficient without additional learnable parameters. Therefore, using SESA can efficiently create effective spectral views and significantly improve detection performance.

2) Effectiveness of Pyramid SSM: The proposed pyramid SSM can capture multiresolution spectral long-range dependencies to enhance feature representation. To verify its effectiveness, we first compare it with the base model without the pyramid SSM and with models using different pyramid

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TABLE V
PERFORMANCE COMPARISON OF DIFFERENT BACKBONES ON FOUR DATASETS

Dataset		MLP	LSTM	ResNet	ViT	Mamba	FPN	U-Net	PVT	Pyramid SSM
San Diego I	Depth	17	2	9	8	11	1	1	1	1
	Params	0.33M	0.32M	0.34M	0.35M	0.34M	0.33M	0.35M	0.34M	0.33M
	FLOPs	0.52G	0.30G	0.45G	0.56G	0.55G	**0.09G**	0.18G	0.11G	0.14G
	$AUC_{(P_f,P_d)}$	0.9996	0.9993	0.9993	0.9993	0.9996	0.9995	0.9993	0.9990	**0.9998**
	Running time	1.2545	1.1646	1.2236	1.4968	2.6463	**1.1246**	1.3964	1.7053	1.6895
San Diego II	Depth	7	1	7	3	4	1	1	1	1
	Params	0.50M	0.49M	0.49M	0.49M	0.49M	0.49M	0.46M	0.51M	0.49M
	FLOPs	1.74G	**0.18G**	0.99G	1.78G	1.62G	0.21G	0.44G	1.03G	1.17G
	$AUC_{(P_f,P_d)}$	0.9968	0.9983	0.9982	0.9988	0.9978	0.9981	0.9973	0.9977	**0.9989**
	Running time	1.3318	1.1476	1.1841	1.5923	2.3258	**1.1304**	1.3647	1.7536	1.7474
Los Angeles	Depth	6	1	5	3	4	1	1	1	1
	Params	0.51M	0.53M	0.51M	0.53M	0.52M	0.53M	0.49M	0.53M	0.51M
	FLOPs	1.63G	**0.20G**	0.82G	1.94G	1.76G	0.23G	0.47G	1.13G	1.27G
	$AUC_{(P_f,P_d)}$	0.9962	0.9909	0.9963	0.9963	0.9932	0.9961	0.9946	0.9948	**0.9976**
	Running time	1.3216	1.1711	1.1715	1.6874	2.3912	**1.1491**	1.4144	1.7761	1.7729
Pavia	Depth	16	2	8	7	10	1	1	1	1
	Params	0.33M	0.37M	0.34M	0.32M	0.33M	0.36M	0.36M	0.35M	0.33M
	FLOPs	0.64G	0.39G	0.53G	0.64G	0.65G	**0.12G**	0.23G	0.15G	0.19G
	$AUC_{(P_f,P_d)}$	0.9770	0.9267	0.9875	0.9657	0.9506	0.9735	0.9901	0.9818	**0.9923**
	Running time	1.6198	1.4063	1.5302	2.0458	4.1172	**1.3485**	2.0392	2.6473	2.7239

TABLE VI
AUC COMPARISON OF COMPETING METHODS ON SENSITIVITY TO TARGET SPECTRAL DIVERSITY ON SAN DIEGO I. BOLD REPRESENTS THE BEST RESULT WHILE UNDERLINED THE SECOND

Spectrum	(9, 88)	(11, 87)	(12, 86)	(11, 88)	(9, 90)	(9, 91)	(15, 90)	(10, 89)
OSP	0.9363	0.9975	0.9979	0.9978	0.8920	0.9049	0.4387	0.9970
CEM	0.9155	0.9877	0.9968	0.9865	0.9836	0.9807	0.9325	0.9966
CSCR	0.9986	0.9986	0.9985	0.9984	0.9984	0.9987	0.9862	0.9982
CTTD	0.8824	0.8740	0.9975	0.8519	0.9017	0.9948	0.8829	0.9978
BLTSC	0.9587	0.9897	0.9980	0.9828	0.9855	0.9715	0.9084	0.9971
MLSN	0.4957	0.9937	0.9927	0.5329	0.9897	0.9660	0.0578	0.9965
OS-VAE	0.9759	0.9935	0.9828	0.9889	0.9855	0.9777	0.9732	0.9912
TSTTD	0.9965	0.9955	0.9964	0.9924	0.9956	0.9973	0.9805	0.9963
HTD-Mamba	**0.9991**	**0.9995**	**0.9996**	**0.9988**	**0.9995**	**0.9988**	**0.9969**	**0.9995**

TABLE VII
AUC COMPARISON OF COMPETING METHODS ON SENSITIVITY TO DIFFERENT LEVELS OF NOISE ON SAN DIEGO I. BOLD REPRESENTS THE BEST RESULT WHILE UNDERLINED THE SECOND

SNR (dB)	5	10	15	20	25	30	35	40
OSP	0.9285	0.9869	0.9927	0.9948	0.9957	0.9964	0.9978	0.9973
CEM	0.9740	0.9917	0.9944	0.9962	0.9981	0.9986	0.9981	0.9985
CSCR	**0.9899**	0.9963	0.9979	0.9983	0.9983	0.9982	0.9983	0.9984
CTTD	0.7058	0.8296	0.9552	0.9906	0.9970	0.9976	0.9978	0.9975
BLTSC	0.9715	0.9937	0.9961	0.9970	0.9983	0.9988	0.9983	0.9986
MLSN	0.5614	0.5975	0.6818	0.8173	0.9485	0.9912	0.9928	0.9948
OS-VAE	0.9725	0.9914	0.9949	0.9962	0.9981	0.9985	0.9979	0.9980
TSTTD	0.9333	0.9842	0.9948	0.9957	0.9962	0.9965	0.9963	0.9971
HTD-Mamba	0.9888	**0.9971**	**0.9987**	**0.9993**	**0.9993**	**0.9995**	**0.9995**	**0.9996**

levels. Table IV shows the comparison results. It can be observed that after removing the pyramid SSM, the detection performance significantly declines, illustrating its effectiveness. Meanwhile, as the level increases from 1 to 4, the $AUC_{(P_f,P_d)}$ scores show an overall upward trend. This demonstrates that the multiresolution spectral features can effectively enhance feature representation and improve detection performance.

To further verify the superiority of the pyramid SSM, we compare it with several mainstream backbones, including MLP [51], long short-term memory (LSTM) [52], residual network (ResNet) [53], vision Transformer (ViT) [54], Mamba [35], feature pyramid network (FPN) [55], U-Net [56], and pyramid ViT (PVT) [57]. Among these methods, MLP, LSTM, ResNet, ViT, and Mamba are single-scale-based models, while FPN, UNet, and PVT are multiscale-based models considering different resolutions. Since ResNet, ViT, FPN, UNet, and PVT are originally based on RGB images, we modified them into the 1-D form for spectral processing. For a fair comparison, the parameter capacity of all the competing models is set to roughly the same by adjusting the network depth. The comparison results are shown in Table V. It can be found that all the competing models achieve promising performance, but our pyramid SSM obtains the best results on all four datasets. In addition, with similar parameters, pyramid SSM requires relatively low floating point operations (FLOPs), and the running time is also within a feasible range for practical applications. Therefore, the proposed pyramid SSM is effective and efficient for extracting discriminative intrinsic features in HTD due to its consideration of multiresolution global correlations and the hardware parallel scanning mechanism.

E. Robustness Analysis

In this section, we evaluate and analyze the robustness of the proposed method, focusing on its sensitivity to spectral diversity and noise.

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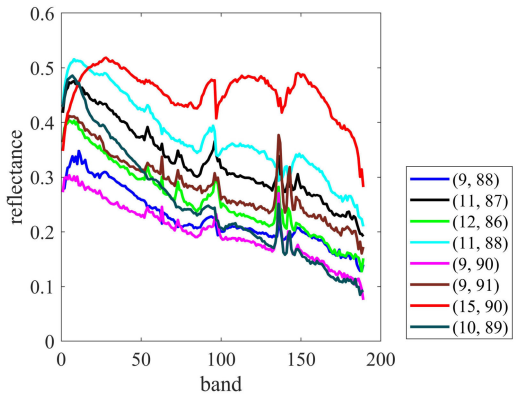


Fig. 7. Spectral curves of the selected target spectra on San Diego I.

1) Sensitivity to Spectral Diversity: To investigate the sensitivity of the proposed method to spectral diversity, we select multiple distinct target spectra, each used individually as prior knowledge. Specifically, we cluster the target pixels based on their spectral features using the k-means algorithm, and select one sample from each resulting cluster, yielding multiple distinct target spectra. These spectra are then used individually as priors. For simplicity and representativeness, we conduct experiments using the San Diego I dataset and set the number of clusters to 8 to explore the sensitivity to target spectra. The selected eight target spectra correspond to the following pixel coordinates: (9, 88), (11, 87), (12, 86), (11, 88), (9, 90), (9, 91), (15, 90), and (10, 89), with their spectral curves displayed in Fig. 7. The quantitative $\text{AUC}_{(P_f, P_d)}$ comparison of competing methods is presented in Table VI.

As shown in Table VI, CSCR and HTD-Mamba outperform other methods in terms of robustness. CSCR combines sparse and collaborative representations to model each testing pixel, preserving robustness by leveraging relationships between pixels and reducing reliance on a single target spectrum. However, its performance can still be affected by spectral diversity. For example, when the target pixel at coordinates (15, 90) is used as the target prior, performance declines due to the large spectral discrepancy from the other target spectra, as shown in 7. In contrast, HTD-Mamba achieves the most stable results across the eight distinct spectra by extracting discriminative intrinsic spectral features from each pixel. Furthermore, the proposed pyramid SSM captures multiresolution global spectral features, reducing the impact of spectral variation and ensuring consistent detection performance. As a result, HTD-Mamba is remarkably robust in target spectral diversity, enhancing its practical applicability.

2) Sensitivity to Noise: To investigate the robustness of HTD-Mamba in noisy environments, we add Gaussian white noise at varying levels to the testing HSI while keeping the target spectrum fixed. Specifically, noise with signal-to-noise ratios (SNRs) ranging from 5 to 40 dB, with a step size of 5 dB, is added to the San Diego I dataset. The quantitative $\text{AUC}_{(P_f, P_d)}$ comparison of competing methods is presented in Table VII.

As shown in Table VII, as the SNR decreases, the detection performance of all competing methods declines. However, our HTD-Mamba method remains only marginally affected by the added noise and consistently outperforms other methods across most noise levels. In scenarios with extremely low SNR, its performance is slightly inferior to that of CSCR. This robustness can be attributed to the ability of HTD-Mamba to extract intrinsic spectral features and fully leverage multiresolution global features. As a result, the proposed method demonstrates strong resilience to noise, making it highly applicable in practical scenarios.

IV. Conclusion

This article introduces a spectrally contrastive learning method based on Mamba for HTD. It addresses the problem of limited training samples by constructing spectral view pairs for contrastive learning and tackles spectral variation by extracting discriminative intrinsic features using a pyramid SSM. Specifically, pixel-level instances are recognized by maximizing the feature similarity of positive view pairs while minimizing that of negative ones. First, we propose an SESA technique to create spectral views while leveraging spatial information, thereby obtaining sufficient view pairs. Then, we transform the view pairs into spectral sequences using group-wise spectral embedding and introduce Mamba to extract global features by capturing long-range dependencies with linear complexity. Furthermore, we develop a pyramid SSM to obtain multiresolution feature representations by extracting the global correlation of spectral sequences with different resolutions and fusing features from different levels. Experiments verify the effectiveness and superiority of the proposed method in terms of quality evaluation and ablation studies.

Despite the promising performance of HTD-Mamba, there is potential for further improvement. The current backbone focuses solely on spectral features, so future work could explore integrating both spectral and spatial Mamba networks to enhance detection robustness. In addition, training and testing on the same dataset may lead to high time consumption across multiple tasks. To address this, we plan to apply transfer learning by pretraining on a public dataset and fine-tuning on others, improving generalization and computational efficiency.

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